

We plan to develop a game using the Java programming language and the libGDX framework. For graphics, we will use png files for the sprites, swapping sprites for the pencil based on the amount of graphite, and create a tile map system for the board. We plan to implement multiple unique levels that progressively get more difficult as the player progresses through them. On the main menu, the player can select the level to play, with each level being locked until they complete the previous one. Our game takes place on a sheet of paper as the board, with a pencil as the main character. The pencil's health is determined by the amount of graphite that it has inside, changing the size of the pencil sprite, and the player loses the game if they run out of graphite. As the player moves, they draw a graphite trail behind them, slowly losing graphite over time during movement and losing large amounts upon contact with an enemy or hazard. The player's goal is to visit each predetermined plot point on the paper, therefore connecting the points to create the final drawing. Additionally, the player can gain points by collecting graphite shards, which heals the pencil and gives extra points when they have reached max health.

There are two different types of moving enemies: the eraser and the pencil sharpener. The eraser is a moving enemy that pathfinds towards the player, and upon contact will deal damage to the player by taking graphite. When the player comes into contact with the pencil sharpener, it will hold the player for a given amount of time, dealing constant damage until the player escapes. They can also escape faster based on the amount of keyboard inputs while they are being held, encouraging button spamming to escape with minimal damage. Additionally, there are two different non-moving hazards: ink and white-out. The ink appears as a stationary blob and if the player comes into contact with the ink, their movement speed will be significantly reduced until they can leave. Moving enemies will not be affected by this speed decrease, providing a great opportunity to damage the player. The white-out puddle will not give any status effects, but will deal damage to the player upon contact, similar to the function of the eraser enemy.

There are multiple rewards that the player can collect throughout the game. The primary win condition reward is the previously mentioned plot points, which the player must visit each one in order to complete the level. Additionally, the player can encounter wood shavings, which reward the user with bonus points and increase the maximum amount of health the player can have. The player can also find lead refill canisters, which will restore a large amount of health instantaneously, however the player only has a limited amount of time to collect this reward before it disappears. Additionally, there will be a time limit that the player must race against to ensure they can connect all plot points before the timer expires. If the timer reaches zero before the player has completed the main task, they will lose and must restart the level. As an extra goal for the player, the total amount of points will be displayed at the end of the level, encouraging them to go out of their way to gain as many bonus points as possible, ensuring they still connect each plot point before the timer expires.